



Icescopy Manual

User guide for image-based freezing analysis

Product manual

Current user documentation for the Icescopy desktop application.

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This manual is written as task-oriented product documentation.

Writing standard used for this manual

- Diataxis for document structure
- Google developer documentation guidance for procedures and prescriptive instructions

What that means in practice

- each page has a clear purpose
- procedures use numbered steps
- action text comes before explanation
- UI names are written consistently
- background detail is kept separate from the main task

Use this reading order

1. Get started
2. Annotate cells
3. Edit images
4. Run analysis, review the Grayscale Plot, and inspect results
5. Import temperature data
6. Manage sessions, export, and preferences
7. Troubleshoot common problems

Scope

- this manual describes how to use the application
- it does not document internal code structure
- developer-facing notes remain outside this folder

1 Get Started

Use this page to load images, create a session, and understand the main workspace.

1.1 Before You Start

Have these ready

- an ordered image sequence
- a writable location for the `.icescopy` session file
- optional external temperature data files if you plan to import them later

1.2 Core Workflow

Use this order for a standard run

1. Create a new session.
2. Add images or add an image folder.
3. Sort the image list if needed.
4. Annotate cells.
5. Assign samples if your experiment uses sample groups.
6. Run analysis.
7. Review the **Grayscale Plot** dock and the **Results Tables** dock.
8. Import external temperature data if needed.
9. Save the session.
10. Export result tables.

1.3 Start A New Session

1. Open **File -> New Session**.
2. Add images with **File -> Add Images...** or **File -> Add Folder...**
3. Check the image order in the **Images** dock.
4. Save the session with **File -> Save Session** or **File -> Save Session As...**

Result

- the session is ready for annotation
- the image list and frame slider are populated

Practical rule

- save once near the start so later relink operations can write back into the same session file automatically

1.4 Open An Existing Session

1. Open **File -> Open Session...**
2. Select the `.icescopy` file.
3. Review the restored image list, cells, samples, and results tables.

If the session opens but images are missing:

1. Open **File -> Relink Images Folder...**
2. Select the folder that contains the session images.

Result

- the saved working state is restored

What is restored

- image paths and image order
- cells and keyframe geometry
- sample catalog entries
- image-edit state
- existing results tables

1.5 Main Workspace

The exact dock arrangement can change, but these panels are the main working areas:

- Images
- Tool Options
- Sample Catalog
- Cells
- Grayscale Plot
- Results Tables
- Console
- frame slider and image viewer

1.6 Use The Viewer And Timeline

The viewer supports

- one-image view
- two-image view
- three-image view

When two or three images are shown, you can switch between:

- left-right layout
- top-down layout

The timeline is used for

- frame navigation
- keyframes
- flags
- changing the visible timeline resolution

Use the smaller timeline-resolution slider when the image sequence is long and you need finer control over the main frame slider.

Working pattern

1. Use the large frame slider to scrub through the sequence.
2. Use the small resolution slider to zoom the visible timeline range when precise frame access is difficult.
3. Use the diamond button to mark keyframes when cell geometry changes across the run.
4. Use the flag button to mark frames that need review.

Important

- keyframes and flags are frame markers only
- keyframes affect interpolated cell geometry
- flags are review markers and are exported through `flag_state` in the grayscale results

1.7 Read The Images Dock

The **Images** dock is also a frame marker view.

Marker meanings

- K: keyframe
- F: flagged frame

Use the **Images** dock to:

- confirm image order
- jump to a specific frame
- review where keyframes and flags are already set

1.8 Main Menus

Use **File** for session and export actions:

- new session
- open session
- save session
- save session as
- add images
- add folder
- relink images folder
- output results

Use **Analysis** for processing:

- run analysis
- import temperature data
- CSU and TAMU importers live under this submenu

Use **Window** to show, hide, and reset docks.

1.9 Sort Images

Use **File -> Sort Images** if the loaded order is wrong.

Common sort methods include

- natural filename
- filename ascending
- filename descending
- created time
- modified time
- EXIF time

If a sort method is not valid for the current session, the app rejects it instead of partially applying it.

Use sorting carefully

- sorting changes frame order
- changed frame order invalidates existing analysis results
- if you sort after annotation or analysis, review the sequence again before continuing

1.10 Keyboard Shortcuts

Common shortcuts

- **A**: Cursor
- **S**: Add Cell
- **G**: Grid Tool
- **E**: Edit Cell
- **D**: Delete Cells
- **Z**: Pan and Zoom
- **Space**: temporary pan while working in another tool
- **Enter**: apply the focused action, such as **Set** for freeze frame or **Apply** for crop
- **Cmd+S** or **Ctrl+S**: save session

1.11 Next Step

After loading images, continue with Annotate cells.

2 Annotate Cells

Use this page to place cells, edit existing cells, assign samples, and review cell records.

2.1 Choose The Right Tool

Use these tools by purpose

- **Cursor:** inspect cells, select cells, assign samples, edit freeze frames
- **Add Cell:** place one cell at a time
- **Grid Tool:** place a repeated array of cells
- **Edit Cell:** move or resize existing cells
- **Delete Cells:** remove cells
- **Pan and Zoom:** navigate the image

2.2 Add One Cell

1. Switch to **Add Cell**.
2. Move the preview circle over the well or droplet.
3. Click to place the cell.
4. Repeat for additional cells.

Use this tool when wells are irregular or when you only need a few cells.

2.3 Add A Grid

1. Switch to **Grid Tool**.
2. Move the preview grid into position.
3. Adjust rows, columns, pitch, and tilt in the current tool settings.
4. Click to place the grid.

Use this tool when the image has a regular layout.

2.4 Edit Existing Cells

1. Switch to **Edit Cell**.
2. Select the cell or group of cells to modify.
3. Drag or resize the selection as needed.

Use **Edit Cell** for geometry only. Do not use it for sample assignment or freeze-frame editing.

Common uses

- recenter circles after placement
- adjust radius to match the actual droplet or well boundary
- correct a full row or group after grid placement

2.5 Use Keyframes When Cells Move

Use keyframes when the droplet positions are not stable across the image sequence.

Typical cases

- camera drift
- tray motion
- droplets shifting position over time
- a layout that is stable at some frames but not all frames

How to use them

1. Go to a frame where the cell positions are correct.
2. Toggle the keyframe marker on that frame with the diamond button near the timeline.
3. Adjust the cells for that frame.
4. Move to a later frame where the positions have drifted.
5. Toggle another keyframe.
6. Adjust the cells again for that frame.

Behavior

- if the current frame is a keyframe, that frame stores its own cell geometry
- between two keyframes, cell positions and radii are interpolated
- outside the keyframe range, the nearest keyframe geometry is used

Practical rule

- use as few keyframes as needed to keep circles centered on the wells or droplets
- add more keyframes only where the drift changes enough that interpolation is no longer accurate

Important

- edits on a keyframe update that keyframe
- edits on a non-keyframe frame do not automatically create a new keyframe
- analysis uses the interpolated geometry across frames

Review method

1. Add a keyframe near the start of the sequence.
2. Add another only when the circles visibly drift off target.
3. Scrub between the two keyframes and check the interpolation.
4. If interpolation is not accurate enough, insert another keyframe between them.

2.6 Use Flags During Review

Use the flag button near the timeline to mark frames that need attention.

Typical uses

- suspicious grayscale behavior
- frames with motion blur
- frames worth revisiting before export

Behavior

- flags appear on the timeline
- `flag_state` is exported in the grayscale results table

2.7 Delete Cells

Use either method

1. Switch to **Delete Cells** and click cells to remove them.

Or

1. Switch to **Cursor**.
2. Select one or more cells.
3. Press **Delete** or **Backspace**.

2.8 Select Cells

Use **Cursor** to select cells in the image.

Selection behavior

- single click selects one cell
- box selection selects multiple cells
- selection in the image and selection in the **Cells** panel stay synchronized

You can also select cells while **Image Edit** is active. In that mode, selection is available for inspection, but image-edit controls remain visible.

2.9 Edit Freeze Frames

1. Switch to **Cursor**.
2. Select a cell.
3. Enter the freeze frame in the **Freeze Frame** field.
4. Press **Set** or press **Enter**.

Result

- the cell record is updated
- the change is logged and undoable
- the **Freeze Events** table is rebuilt from the current freeze-frame assignments

Use this when

- the automatic detector missed a real freezing event
- the detector called the wrong frame
- a cell should be left without a freeze event

2.10 Assign Samples

1. Switch to **Cursor**.
2. Select one or more cells.
3. Choose a sample in the **Sample ID** control.

To create a new sample from the current context:

1. Use **New Sample**.

Autogenerated sample names follow the pattern defined in Preferences.

2.11 Use The Sample Catalog

Use the **Sample Catalog** dock to manage the session-level sample list.

Current behavior

- sample IDs are fixed numeric labels
- sample names are editable in the table
- sample assignment still happens in **Cursor**

Use it to

- rename samples
- add unused samples ahead of time
- review the current sample list

Restriction

- a sample cannot be deleted while any cells are still assigned to it

Why it matters

- CSU temperature import matches external sample columns against current sample names
- clear sample naming before import reduces manual cleanup later

2.12 Use The Cells Panel

The **Cells** panel is a read-only inspection and selection view.

It shows

- Cell ID
- Sample ID
- Sample Name
- Freeze Frame(s)

Use it to

- review cell records
- select cells from a list
- compare cell metadata quickly

Do not use it for editing. Editing stays in **Cursor**.

2.13 Recommended Annotation Order

For most runs, use this order:

1. Add or grid the initial cells.
2. Clean up geometry with **Edit Cell**.
3. Add keyframes only if the image sequence drifts.
4. Assign samples in **Cursor**.
5. Rename samples in **Sample Catalog** if needed.
6. Run analysis.

2.14 Next Step

After annotating cells, continue with Edit images or Run analysis, review the Grayscale Plot, and inspect results.

3 Edit Images

Use **Image Edit** to change the image data used for both display and analysis. These edits apply to the whole image, not to individual cells.

3.1 Available Controls

Image Edit includes:

- histogram
- exposure
- contrast
- crop
- uniform exposure

3.2 Review The Histogram

Use the histogram to inspect grayscale distribution before or after editing.

Behavior

- the full-image histogram is shown in gray
- if cells are selected, their histogram is overlaid separately
- if no cells are selected, no red overlay is shown

The histogram is a review tool. It does not change the image by itself.

Use it to answer

- whether the full image is clipped too dark or too bright
- whether selected cells occupy a narrower grayscale range than the full image
- whether exposure or contrast changes are separating droplet signal from background

3.3 Adjust Exposure

1. Switch to **Image Edit**.
2. Move the **Exposure** slider or edit the numeric value.

Use exposure when the full sequence is too dark or too bright.

Behavior

- exposure changes are applied to display and analysis
- each committed change is undoable

3.4 Adjust Contrast

1. Switch to **Image Edit**.
2. Move the **Contrast** slider or edit the numeric value.

Use contrast when wells or droplets are difficult to distinguish.

Behavior

- contrast changes are applied to display and analysis
- each committed change is undoable

3.5 Crop The Image

1. Switch to **Image Edit**.
2. Click **Crop**.

3. Adjust the crop box.
4. Click **Apply** or press **Enter**.

To discard the current crop draft

1. Click **Cancel**.

To remove the committed crop

1. Click **Reset**.

Behavior

- crop does not start automatically when you enter **Image Edit**
- crop is committed only when you apply it
- crop changes are logged individually for undo and redo

3.6 Run Uniform Exposure

Use uniform exposure when brightness drifts from frame to frame.

1. Switch to **Image Edit**.
2. Set the uniform-exposure area on a stable reference region.
3. Run uniform exposure.

Choose an area that is

- stable across frames
- representative of the illumination you want to normalize
- not dominated by changing ice or fluid behavior

Behavior

- correction is per image
- correction affects display and grayscale analysis
- the current frame is used as the reference frame
- **Run** and **Reset** are both undoable

Use a control area that is

- stable across the sequence
- representative of the illumination drift
- not dominated by changing freezing behavior

3.7 Select Cells During Image Edit

You can still select cells while **Image Edit** is active.

Use that when you want to

- compare histograms for selected cells
- inspect a subset of the image while staying in image-edit mode

Exception

- while you are actively manipulating a crop draft, crop takes priority and selection is suppressed

3.8 What Image Edit Affects

Image-edit settings affect

- image display

- grayscale analysis
- temperature-sync outputs that depend on the current freeze results

Image Edit does not move cells. Cell geometry still belongs to the annotation tools and keyframe workflow.

Practical rule

- use **Image Edit** to improve signal quality
- use annotation tools and keyframes to improve geometric tracking
- these are separate workflows and should stay separate

3.9 Undo And Redo

These image-edit actions are tracked individually:

- exposure changes
- contrast changes
- run uniform exposure
- crop apply
- crop reset
- uniform exposure reset

Current behavior

- image-edit undo and redo preserve the existing results tables in the session state
- preserved tables are not automatically recomputed
- if the edits change the interpretation of freezing, run analysis again

3.10 Next Step

After image edits are in place, continue with Run analysis, review the Grayscale Plot, and inspect results.

4 Run Analysis, Review The Grayscale Plot, And Inspect Results

Use this page to run grayscale analysis, inspect result tables, and export data.

4.1 Run Analysis

1. Load images.
2. Annotate cells.
3. Apply any required image edits.
4. Open **Analysis** -> **Run Analysis**.

Result

- the app processes the loaded image sequence frame by frame
- result tables are refreshed from the current session state

Analysis uses

- cell positions
- sample assignments
- image-edit adjustments

If cells move across the sequence

1. Add keyframes before running analysis.
2. Adjust cell geometry at the key frames.
3. Let the app interpolate between them.

If keyframes are not used, analysis uses the current static cell geometry for all frames.

Before you run

- confirm image order
- confirm circles are centered
- add keyframes if the layout drifts
- apply any exposure, contrast, crop, or uniform-exposure edits first
- save the session if the run represents a major milestone

4.2 Know What Invalidates Analysis

These operations invalidate analysis results:

- adding images
- removing images
- sorting or reordering images

These operations can preserve existing tables in session state, but do not recompute them automatically:

- cell edits
- image edits
- sample edits

Practical rule

- if the interpretation of the data changed, run analysis again

4.3 Review Result Tables

The **Results Tables** dock contains three tabs:

- **Measurements**
- **Freeze Events**
- **Temperature Sync**

Use them by purpose

- **Measurements** for frame-by-frame grayscale and geometry output
- **Freeze Events** for per-cell event indexing
- **Temperature Sync** for time and temperature alignment after external import

4.4 Measurements

Use **Measurements** in **Results Tables** for per-frame grayscale values. This table is the main frame-based output from analysis.

Current column structure

- `file_name`
- `flag_state`
- `cell_<id>_grayscale`
- `cell_<id>_circle_x`
- `cell_<id>_circle_y`
- `cell_<id>_circle_radius`

Typical uses

- inspect grayscale values by frame
- review flagged frames
- export frame-based data to CSV

Important

- this table does not depend on EXIF time
- time and temperature are added later through temperature import

4.5 Freeze Events

Use **Freeze Events** to review the image index where freezing was recorded.

Current columns

- `cell`
- `image_index`
- `image_name`

This table is intentionally image-based. It does not mix in external time or temperature assumptions. Manual freeze-frame edits also update this table.

4.6 Temperature Sync

Use **Temperature Sync** after a successful temperature import.

This table combines

- image-derived freeze information
- external time and temperature data

Do not expect this tab to appear before temperature import.

4.7 Review The Grayscale Plot

Use the separate **Grayscale Plot** dock to inspect traces across frames.

Typical checks

- confirm that the grayscale trace looks reasonable
- compare selected cells
- inspect the convolution trace against the raw trace
- verify freeze events against the image sequence

Use the plot for two different tasks

- checking whether the grayscale signal itself is sensible
- checking whether the detector is calling the right freezing frames

Read the traces this way

- the solid trace is the raw mean grayscale through time
- the dashed trace is the convolution-based edge-enhanced signal used to find sudden drops
- a strong freezing event usually looks like a clear grayscale drop with a corresponding response in the dashed trace

Use the plot together with the image viewer:

1. Find a suspicious dip or peak in the plot.
2. Scrub to the corresponding image frames.
3. Decide whether the detector is responding to real freezing, noise, blur, or drift.

4.8 Tune Freeze Finding

Freeze-finding parameters live in

- Preferences -> Analysis -> Freeze Finding

Use the **Grayscale Plot** and the **Freeze Events** table together when tuning.

Recommended order

1. Start with **Peak Width**.
2. Then adjust **Peak Prominence**.
3. Only after that, adjust the convolution settings if needed.

Parameter guide

- **Peak Width**

Increase it to reject narrow noisy dips. Decrease it if real freezing events are being missed because the detected dip is too narrow.

- **Peak Prominence**

Increase it to require a stronger event. Decrease it if obvious freezing dips are not being accepted.

- **Tail Extension Points**

Increase it if freezing happens near the end of the run and end-of-trace events are being missed.

- **Convolution Half Window Points**

Controls the width of the step kernel used to build the dashed convolution trace. Larger values make the detector respond to broader changes. 0 keeps the original whole-trace window behavior.

- **Convolution Ramp Points**

Softens the step kernel near the center. Increase it if the real grayscale drop is gradual or sloped rather than abrupt.

Tune by failure mode

- too many false events:

increase **Peak Prominence** first, then increase **Peak Width**

- real freezing missed:

decrease **Peak Prominence** first, then decrease **Peak Width**

- late-run events missed near the end:

increase **Tail Extension Points**

- dashed trace reacts to very broad trends instead of sharp drops:

reduce **Convolution Half Window Points**

- dashed trace is too sharp for a gradual freeze transition:

increase **Convolution Ramp Points**

Practical workflow

1. Run analysis with the current settings.
2. Inspect the raw trace and dashed convolution trace in **Grayscale Plot**.
3. Compare reported freeze events with the image sequence.
4. Adjust one parameter at a time.
5. Run analysis again and compare the result.

Do not tune by table output alone. The plot is the faster way to see whether the convolution trace is reacting to real freezing or to noise.

Use a controlled tuning process

1. Pick one representative cell that behaves clearly.
2. Tune one parameter at a time.
3. Re-run analysis after each change.
4. Check a few easy cells and a few difficult cells before accepting the new settings.

4.9 Export Result Tables

1. Open **File -> Output Results**.
2. Choose **Grayscale CSV...**, **Freeze CSV...**, **Temperature CSV...**, or **All CSVs...**
3. Save the CSV file.

Available exports depend on the current session contents.

4.10 Interpretation Notes

Use these assumptions when reading the tables:

- grayscale analysis is frame-based
- image-edit settings affect analysis, not just display
- temperature-aware interpretation belongs in the **Temperature Sync** table, not in the core frame table

4.11 Next Step

If you need time and temperature alignment, continue with Import temperature data.

5 Import Temperature Data

Use this page to align image-derived freezing results with external temperature data.

[Open temperature import from](#)

- **Analysis -> Import Temperature Data**

5.1 Choose The Correct Importer

Use

- **CSU IS .dat import...** for CSU data files
- **TAMU Linkam .xlsx import...** for TAMU Linkam workbooks

The core analysis stays frame-based. These importers add time and temperature later.

5.2 Import CSU Data

Use the CSU importer when the external **.dat** file contains the authoritative image-to-time mapping.

1. Open **Analysis -> Import Temperature Data -> CSU IS .dat import...**
2. Select the CSU **.dat** file.
3. Select any blank samples if blank correction is required.
4. Set **Reset After Warmed To (°C)** if the run contains repeated cooling and warming cycles.
5. Confirm the import.

Current CSU behavior

- uses the **Picture** column to match images to temperature rows
- matches samples by sample name
- reports unmatched names
- corrects both false negatives and false positives relative to the current image-based freeze results
- allows blank-sample selection for CSU blank correction

Interpretation

- the CSU **.dat** file is the authoritative time and temperature source
- the app trusts the **.dat** picture-to-row association instead of EXIF time or file modified time
- if a sample name does not match, that sample is reported instead of silently guessed

5.3 Import TAMU Data

Use the TAMU importer when image timestamps are encoded in filenames and temperature comes from a Linkam workbook.

1. Open **Analysis -> Import Temperature Data -> TAMU Linkam .xlsx import...**
2. Select the workbook.
3. Optionally select the calibration CSV.
4. Set **Reset After Warmed To (°C)** if the run contains repeated cooling and warming cycles.
5. Confirm the import.

Current TAMU behavior

- reads image time from the PNG filename
- interpolates temperature from the Linkam temperature trace
- optionally applies well-level calibration by cell ID
- if no sample setup exists, all cells are treated as one output group
- builds frozen counts from Icescopy freeze results

Interpretation

- the workbook provides the temperature trace
- the image filename provides the timestamp anchor for each frame
- temperature is interpolated onto the image times before the output table is built

5.4 Use Reset After Warmed To

Use **Reset After Warmed To** (°C) when the experiment contains repeated cooling and warming cycles.

Meaning

- once temperature warms back to the selected threshold, the next cooling segment is treated as a new cycle
- counts are interpreted within each cycle instead of across the entire run
- crossing the threshold counts; the trace does not need to hit the exact value

The cycle detector also uses the analysis preference **Cycle Warm-Up Hysteresis** (°C) to ignore tiny threshold jitter.

Choose this threshold deliberately

- set it high enough that a genuine warm-up must occur before a new cycle starts
- do not set it so low that tiny oscillations near the threshold create false cycle boundaries

5.5 Understand Blank Correction

Blank correction applies only to the CSU workflow.

Use it when selected samples should act as blanks.

No blank-correction path is used for TAMU.

Current blank-correction rule

- if a selected blank freezes at a CSU row, that blank event contributes to the CSU blank-correction logic from that point onward

5.6 Review The Output

After import, review the **Temperature Sync** tab.

Typical columns include

- `timestamp`
- `temperature_C`
- `cycle`
- `image_name` or `picture`
- optional corrected temperature columns
- per-sample frozen counts

Use this table for downstream work such as:

- frozen count versus time
- frozen count versus temperature
- cycle-by-cycle interpretation for repeated freeze-thaw experiments

5.7 Interpret Repeated Cycles

For repeated freeze-thaw runs

- use the `cycle` column
- compute within-cycle quantities first
- combine cycles later in downstream analysis if needed

Do not treat all cycles as one monotonic run unless that is physically valid for the experiment.

6 Manage Sessions, Export, And Preferences

Use this page to save work, relocate images, export tables, and change application defaults.

6.1 Save A Session

Use **File -> Save Session** to write the current session.

Behavior

- if the session already has a file path, **Save Session** updates that file
- if the session has never been saved, **Save Session** behaves like **Save Session As...**

Use **Cmd+S** or **Ctrl+S** to save quickly.

6.2 Save To A New Location

1. Open **File -> Save Session As...**
2. Choose the destination and file name.
3. Save the **.icescopy** file.

Use this when you want

- a copy in a different location
- a renamed session
- a branch of the current work

6.3 Open A Session

1. Open **File -> Open Session...**
2. Select the **.icescopy** file.

The session restores

- image paths and order
- cells
- sample catalog
- image-edit state
- result tables

Use this to resume work from

- an incomplete annotation pass
- a review session after analysis
- a previously imported temperature-sync session

6.4 Relink An Image Folder

Use this when image paths in a session no longer resolve.

1. Open the session.
2. Open **File -> Relink Images Folder...**
3. Select the folder that contains the session images.

Current behavior

- matches files by basename
- updates the session image paths
- refreshes the current display

- persists the new paths back into the open session file when possible

If the session was opened from an existing `.icescopy` file, relink writes the new paths back into that same file automatically.

Limit

- relink is basename-based
- duplicate filenames in the chosen folder can lead to ambiguous matches

6.5 Export Results

1. Open **File -> Output Results**.
2. Choose **Grayscale CSV...**, **Freeze CSV...**, **Temperature CSV...**, or **All CSVs...**
3. Save the CSV file.

Available exports depend on what exists in the current session.

6.6 Change Preferences

1. Open **Preferences**.
2. Select a category.
3. Change the required values.
4. Click **Save**.

Categories

- **General**
- **Viewer**
- **Drawing**
- **Analysis**
- **Timeseries**
- **Timeline**

Use them by purpose

- **General** for undo, sorting, and sample-name generation
- **Viewer** for image-count and zoom behavior
- **Drawing** for circle and grid defaults
- **Analysis** for freeze finding and cycle interpretation
- **Timeseries** for plot colors and line widths
- **Timeline** for timeline density and marker display

Recommended workflow

- set circle and grid defaults before heavy annotation
- set freeze-finding and cycle preferences before repeated analysis tuning
- set timeline and plot styling only after the core workflow is stable

6.7 Important Preferences

Common settings include

- default circle radius
- default grid parameters
- sample name pattern
- freeze-finder parameters
- cycle warm-up hysteresis

- timeline density
- plot line widths

6.8 Default Sort

The default sort preference controls how newly added images are ordered before you manually re-sort them. Use it when your image sets usually arrive ordered by:

- filename
- filesystem time
- EXIF time

6.9 Sample Name Pattern

Autogenerated sample names use a pattern.

Pattern rules

- # inserts the sample number
- \# inserts a literal #
- \\ inserts a literal backslash

Example

- Sample_# produces Sample_0, Sample_1, and so on

6.10 Preferences File

Application preferences are stored in

- preferences.xml

This file is global application configuration. It is not part of an individual session file.

7 Troubleshoot Common Problems

Use this page to resolve the most common session, packaging, and import problems.

7.1 Session Opens But Images Are Missing

Symptom

- the session opens
- the image list is present
- the images do not resolve

Action

1. Open File -> Relink Images Folder...
2. Select the folder that contains the session images.

7.2 A Session Was Shared With Another Computer

Cause

- the saved absolute image paths do not exist on the second machine

Action

1. Open the session.
2. Relink the image folder on the new machine.

7.3 The Built App Does Not Launch

Cause

- you are running the intermediate PyInstaller build/ output instead of the packaged app

Do not run

- build/Icescopy/Icescopy

Run one of these instead

- dist/Icescopy.app
- dist/Icescopy.app/Contents/MacOS/Icescopy
- dist/Icescopy/Icescopy

7.4 Packaging Fails With Cache Or Permission Errors

Use temporary cache locations during build:

```
PYINSTALLER_CONFIG_DIR=/tmp/icescopy-pyinstaller \  
MPLCONFIGDIR=/tmp/icescopy-mpl \  
XDG_CACHE_HOME=/tmp \  
python -m PyInstaller --clean --noconfirm Icescopy.spec
```

7.5 Temperature Import Looks Wrong

Check these first

- the correct importer was used
- sample names match where required
- the reset temperature is appropriate for the run
- cycle warm-up hysteresis is not too small

- current freeze annotations are correct

Remember

- CSU uses the `.dat` file as the authoritative picture-to-time mapping
- TAMU uses filename time plus interpolation from the Linkam trace

Also check

- whether sample names were finalized before CSU import
- whether blank samples were selected only for CSU
- whether the current freeze frames are correct before import

7.6 A New Cycle Appears At The Wrong Time

Cause

- reset detection depends on both **Reset After Warmed To** (`řC`) and **Cycle Warm-Up Hysteresis** (`řC`)

Action

1. Review the chosen reset temperature.
2. Increase cycle warm-up hysteresis if minor temperature jitter is creating false cycle boundaries.

7.7 Result Tables Look Stale

Cause

- the session state changed after the table was generated

Action

1. Re-run analysis after changing cells, freeze annotations, samples, or image-edit settings.
2. Re-run temperature import after changing freeze annotations or sample assignments.

Notes

- **Cell Edit** and **Image Edit** can preserve existing tables in the session state
- preserved tables are not automatically recomputed
- image add, image removal, and image reorder operations invalidate analysis results

7.8 Cells Drift Off The Droplets During The Run

Cause

- the geometry changes across frames but only one static cell layout is defined

Action

1. Add keyframes at frames where the layout visibly changes.
2. Edit cell geometry on those keyframes.
3. Scrub between keyframes and confirm the interpolation is accurate.
4. Re-run analysis after the geometry is corrected.

7.9 Freeze Finding Misses Real Events Or Calls Too Many

Cause

- the detector parameters do not match the signal shape in the current data

Action

1. Use the **Grayscale Plot** instead of guessing from the tables alone.
2. Compare the solid grayscale trace and dashed convolution trace against the image sequence.
3. Adjust **Peak Prominence** and **Peak Width** first.
4. Adjust convolution parameters only after the basic event thresholding is close.

7.10 The Packaged App Is Large

Cause

- the application bundles PySide6, OpenCV, NumPy, Pandas, SciPy, and PyQtGraph

Result

- a large bundle size is expected unless dependencies are reduced